

# Artificial Intelligence for Business Value

# What is this talk about? (and what it is not!)

## ***What is this talk about:***

- Understand how business decisions are made and the relevance of AI / ML
- Understand the basics of Artificial Intelligence and its relevance in business context
- Discuss industry applications of AI / ML

## ***What this talk is not:***

- Does not deal with cost-benefit analysis of use cases
- Does not cover moral, ethical dimensions of applications
- Does not cover any math behind the techniques

***How delivered:*** I am going to put myself in your shoes, ask & answer key questions that you might have in your mind as you embark on this course!

Q1: What is the motivation for us to understand Data  
Science, AI & ML?



# AI/ML is all around us: Movies at the Theatre (An Example)

## Recommended for you



Open a movie app to  
book a ticket

Watch a movie in  
theatre

Snacks during the  
break

Write a review about  
the movie

Recommendation  
Engine based on  
Collaborative Filtering

Self driving  
car

Predicting box office  
sales based on star  
value, buzz etc.

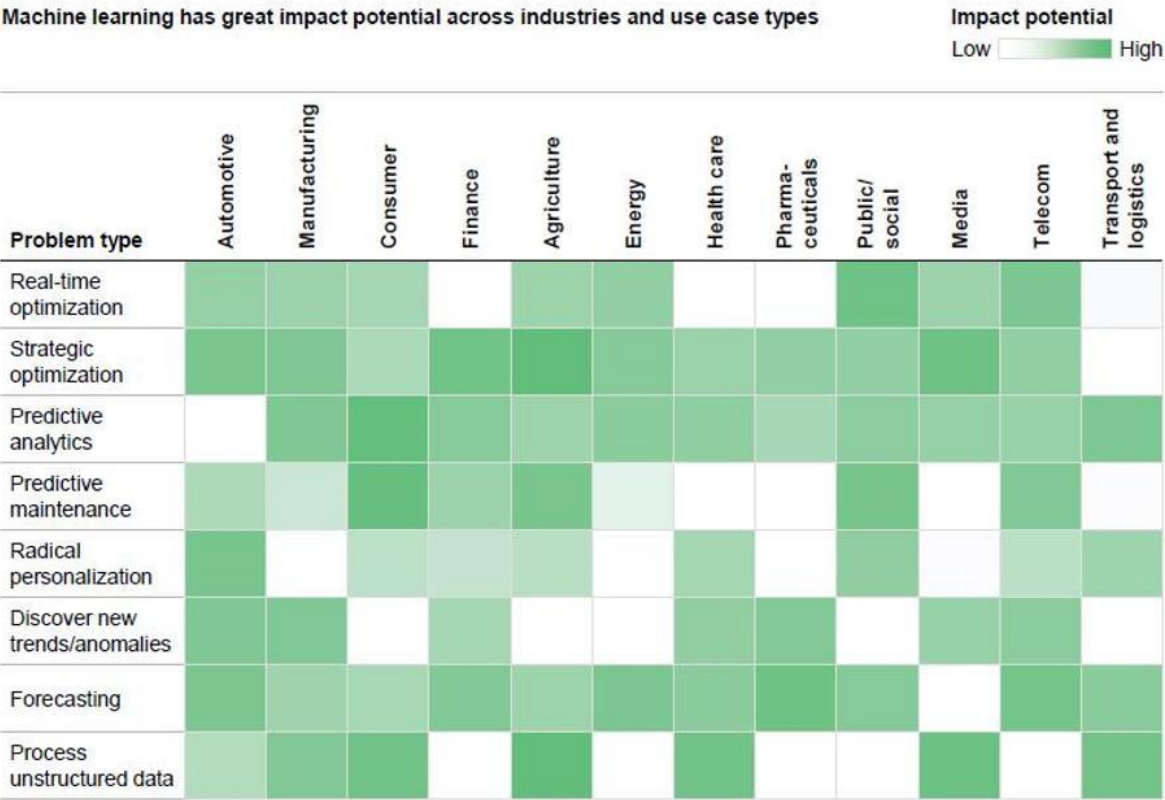
Forecasting product  
sales based on  
historical trends

Text Analytics to  
identify sentiments,  
emotions, likes &  
dislikes etc.

**AI / ML Applications**

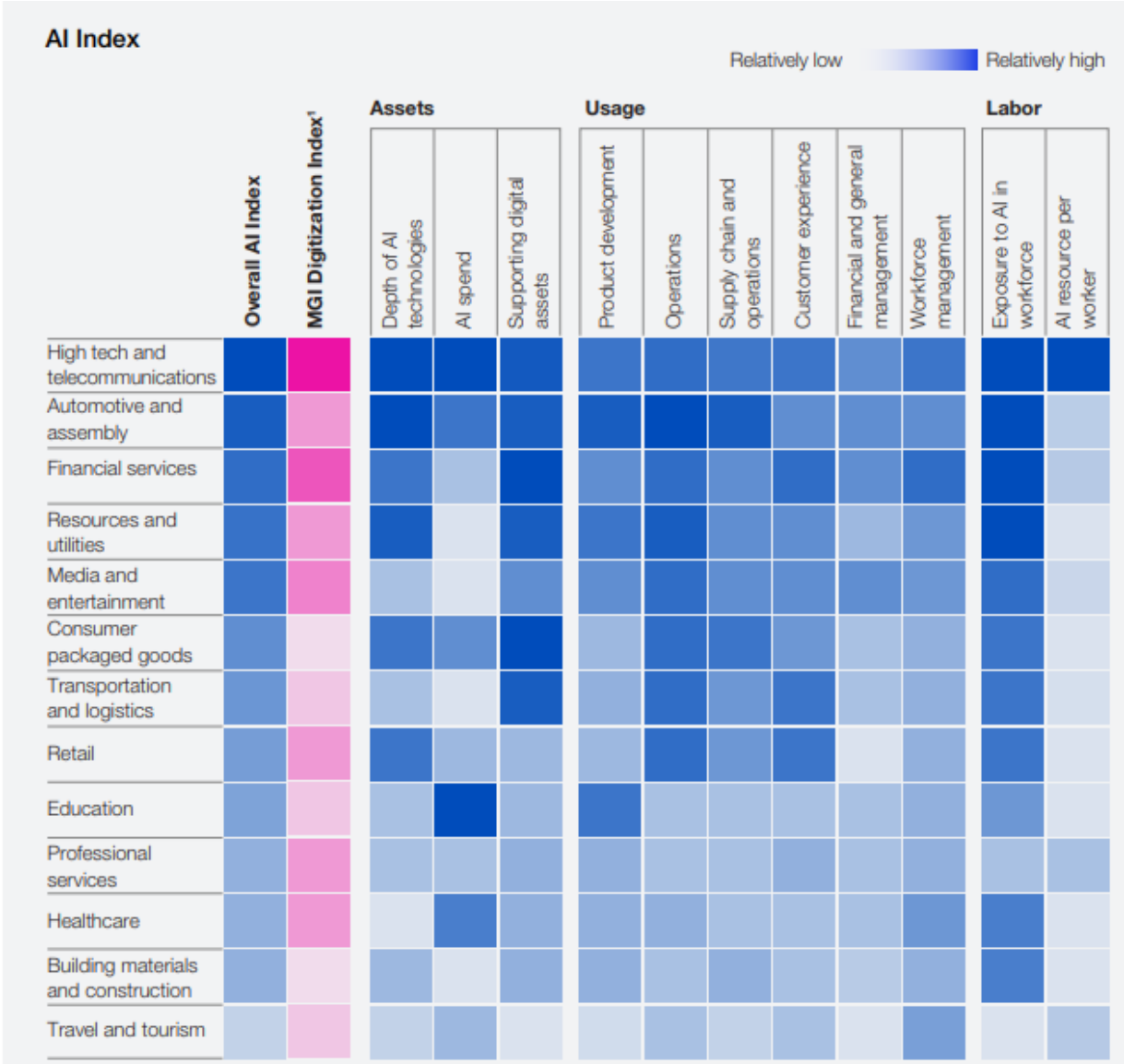
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# There is increasing use of AI/ML across industries = Rewarding Career



SOURCE: McKinsey Global Institute analysis

<https://www.mckinsey.com/business-functions/mckinsey-analytics/our-insights/an-executives-guide-to-ai>



# But getting true business value needs (lot) more work...

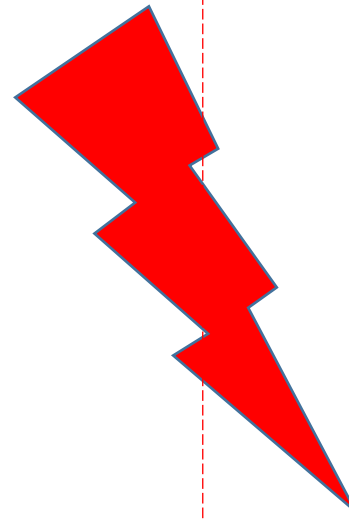
Stacked Ensemble  
has an accuracy of  
97% without  
overfitting

DL Network with 25  
hidden layers beat  
the existing  
benchmarks



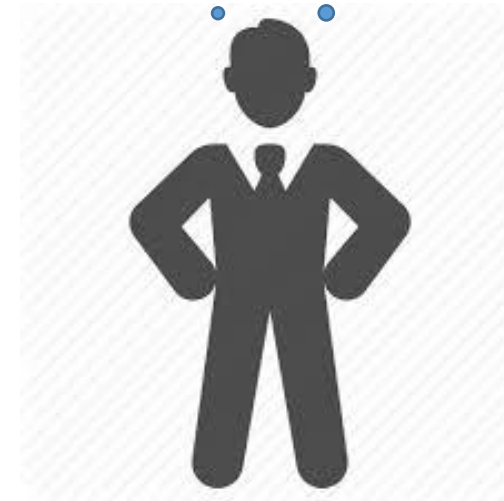
**Data Scientists**

**Data Science (AI/ML) is powerful  
but there exists a divide**



Business Value??  
Not so sure

Do they understand  
the business  
complexity, the  
context?

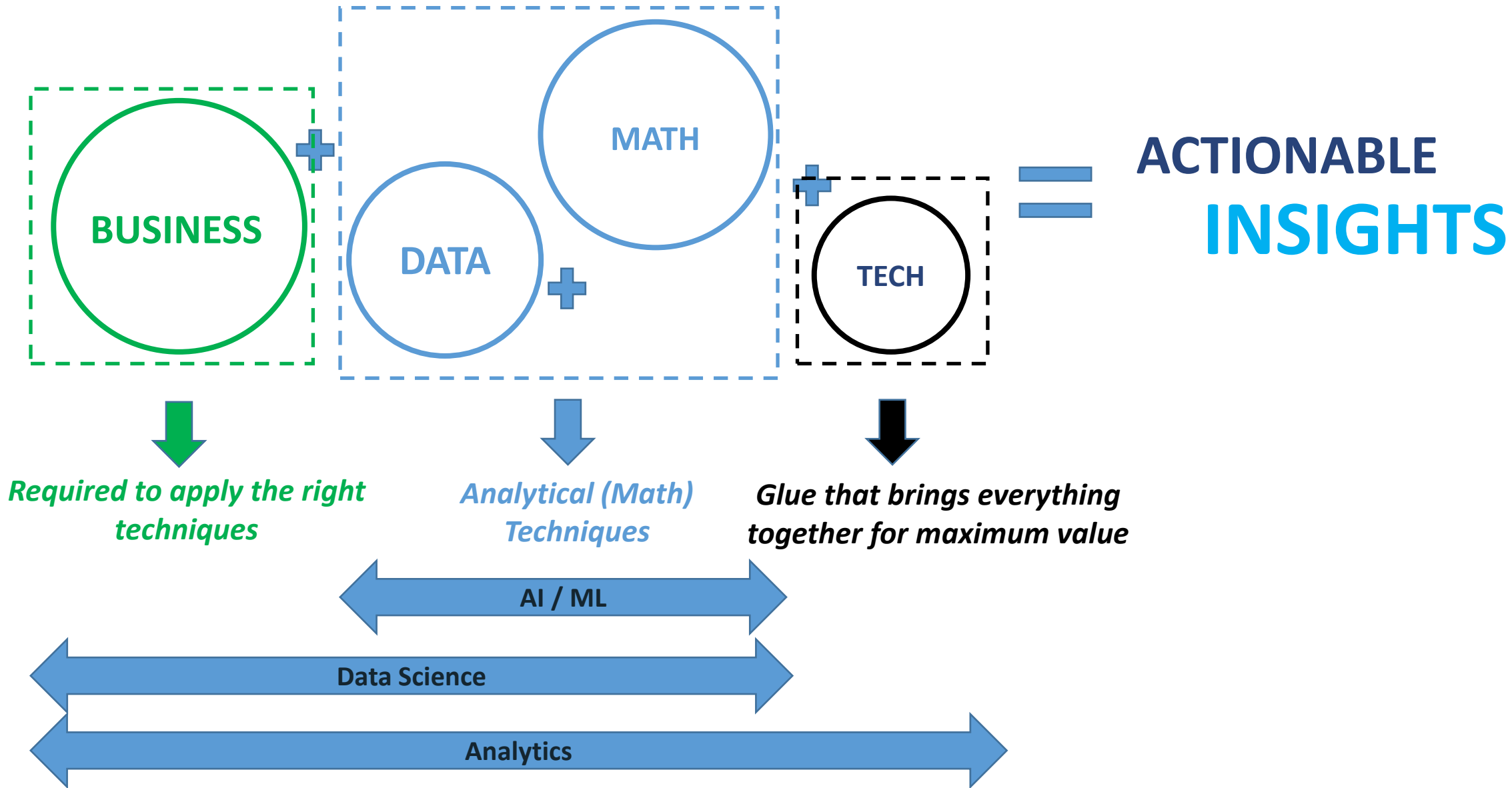


**Business Decision Maker**

Q2: Before getting into the details, can you show me the simplest possible picture to understand Analytics and its relationship to AI / ML?



# Analytics is a tool to solve business problems...





Q3. To start with, can we understand what is Artificial Intelligence? How are AI & ML related?



# First, let's try to understand categories of Human Intelligence



## Intelligence exhibited by this person

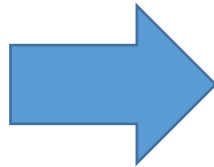
1. Perceive the world, detect signals and collect data
2. Make sense of the world using data (Inference)
3. Decide on the next course of action
4. Act in the Real World

# What is Artificial Intelligence?

Artificial Intelligence refers to the theory and development of computer systems & machines with the ability to perform tasks normally requiring human intelligence

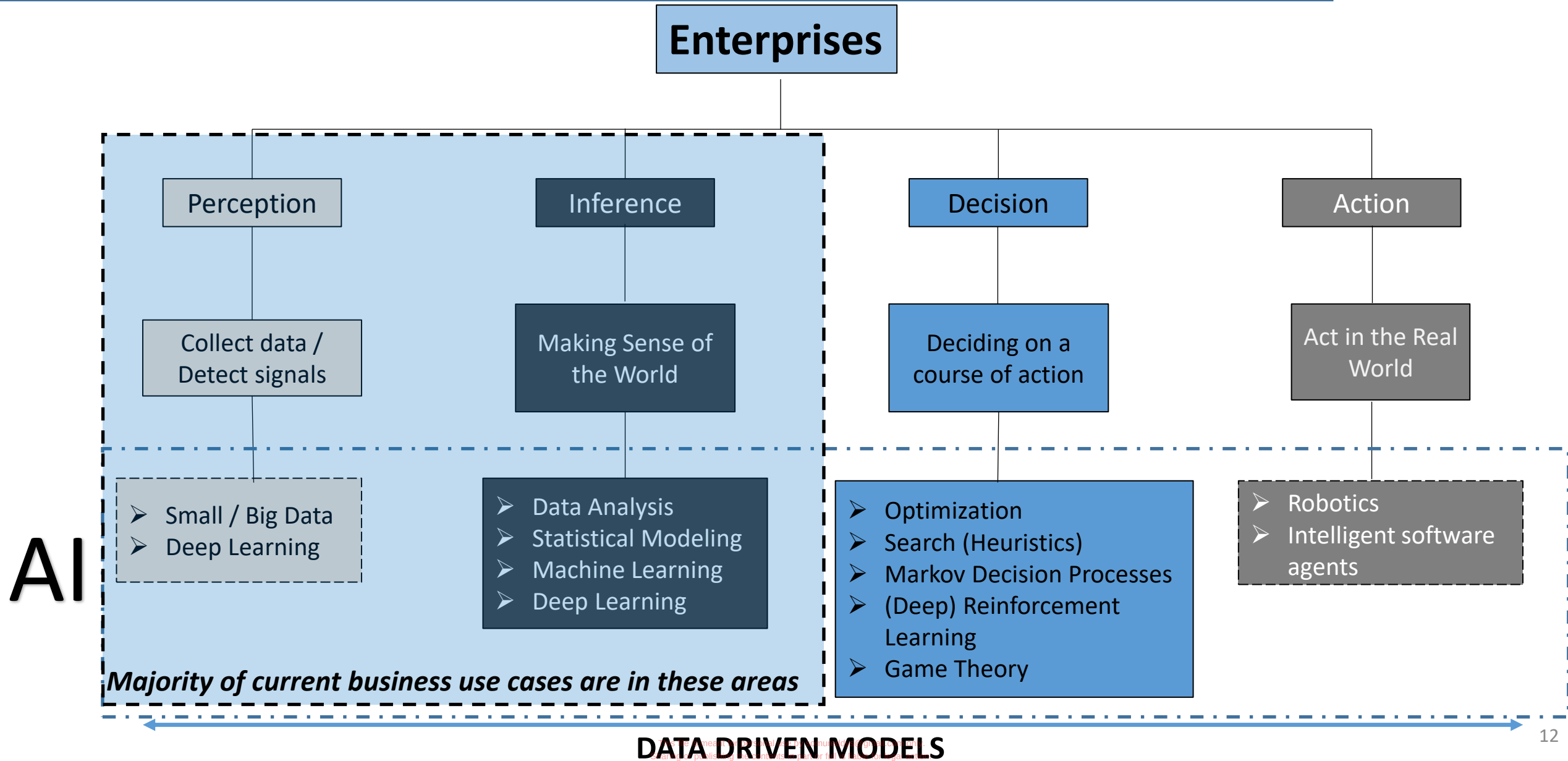
## Main Categories of Human Intelligence

1. PERCEPTION
2. INFERENCE
3. DECISION
4. ACTION

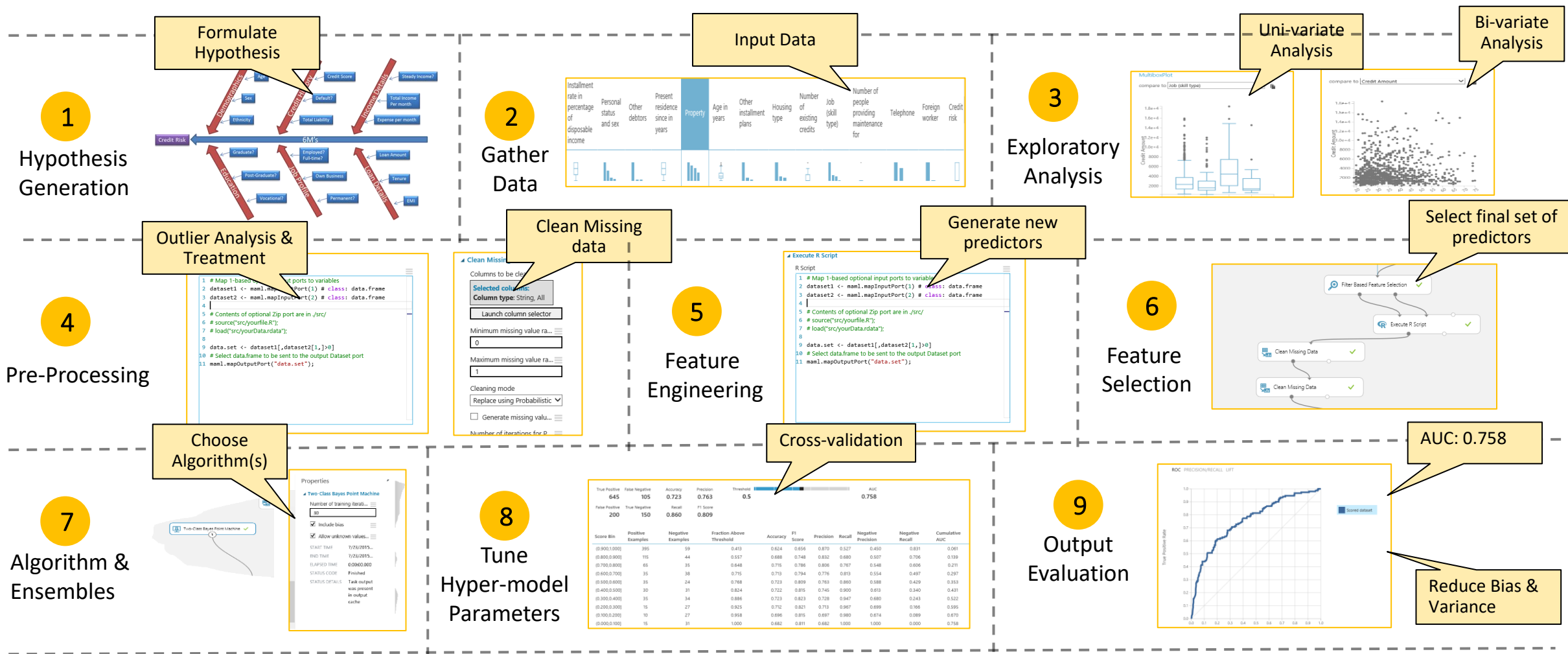


We get our computers to do any or some or all of these tasks by embedding AI into our software programs

# Why is AI relevant for business enterprises?



# Machine Learning Pipeline – Subset of AI



***MANY USE CASES IN THE INDUSTRY ARE BASED ON THIS PIPELINE...***

Q4: What are the broad categories of decisions taken in organizations and how are these decisions made?



# On a daily basis organizations take hundreds of decisions...

Should I acquire Company A?

What revenue guidance should I give my investors?

What price should I set for my product?

How do I increase checkouts on my ecommerce website?

What campaigns should I run? How much should I spend on those campaigns?

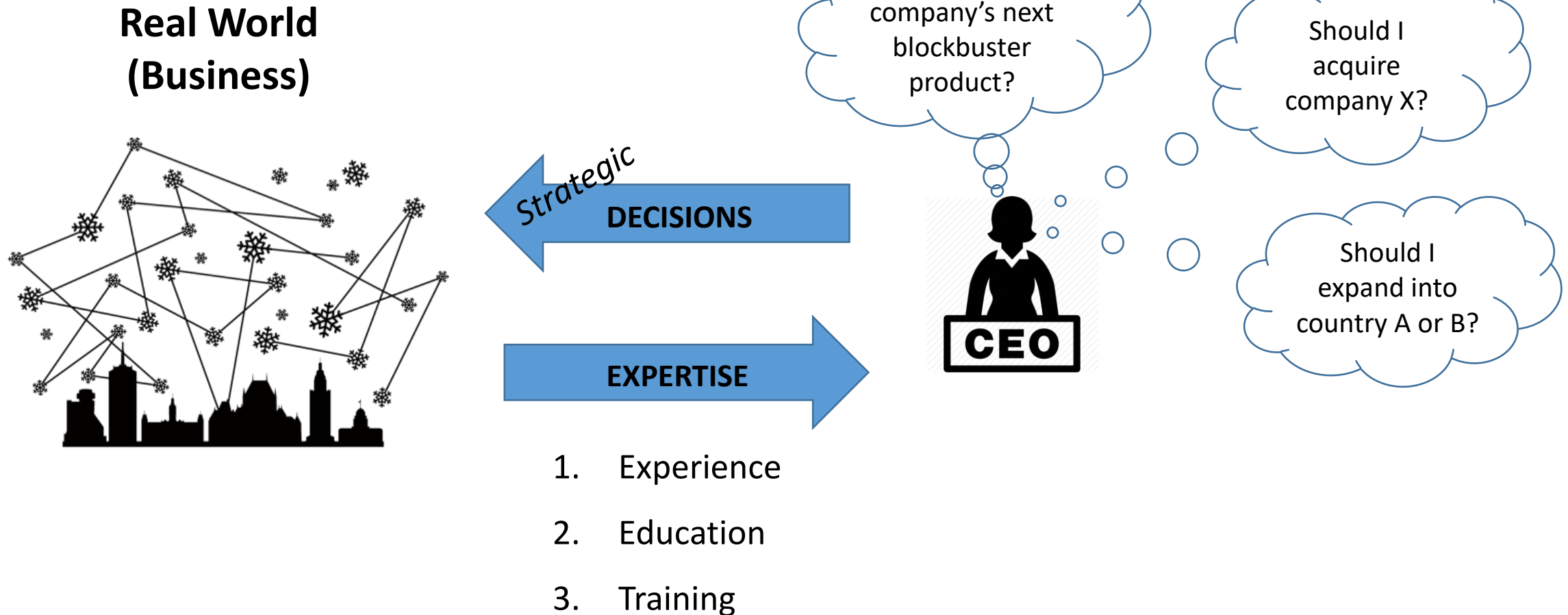
How do I increase manufacturing throughput?

Should I hire this person or not?

What offers should I give my customers?

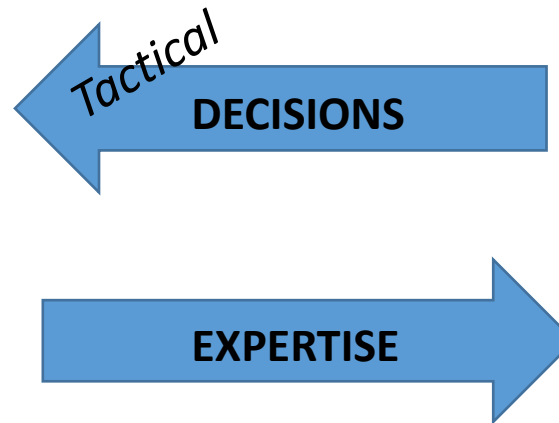
How do I comply with regulatory requirements?

# Business Decisions – Type 1 - Strategic





# Business Decisions – Type 2 - Tactical



1. Education
2. Experience
3. Training



*Head of Sales  
& Marketing*

What price  
should I set for  
Product X, SKU Y?

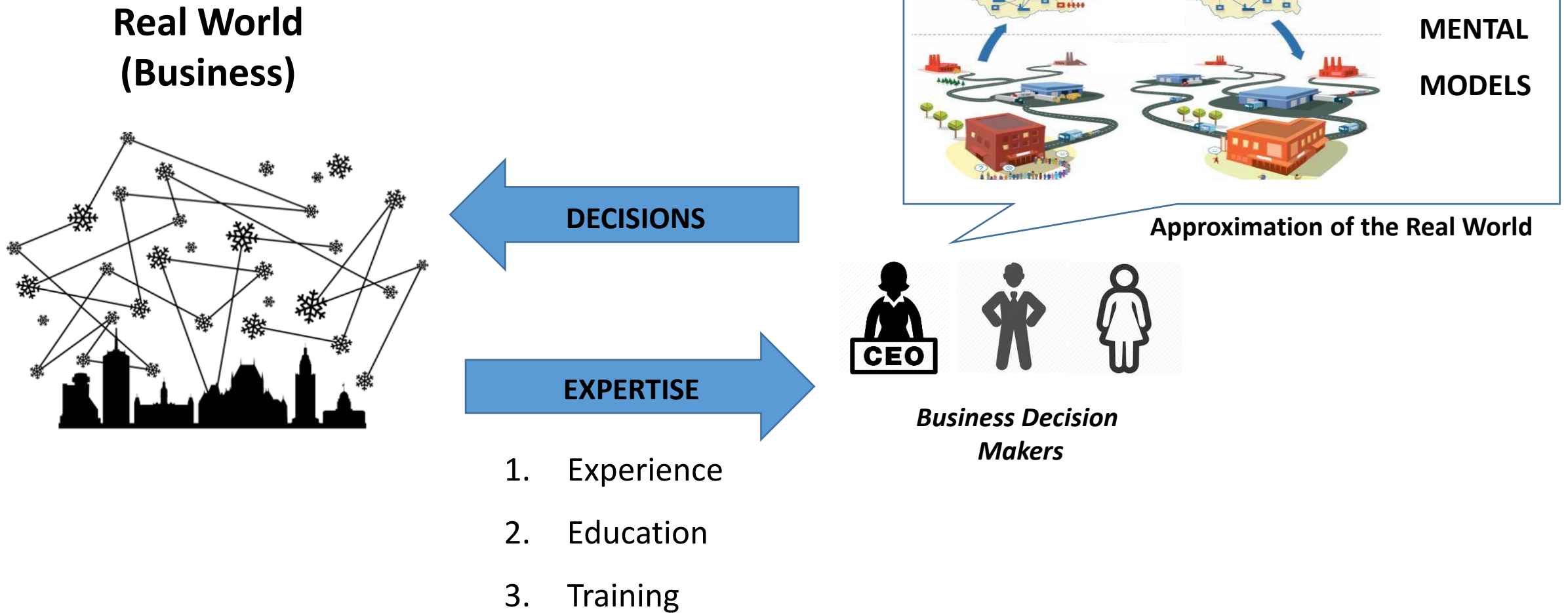
What should the  
next week  
marketing  
campaign focus  
on?

How do I handle  
the negative  
brand perception  
on social media?

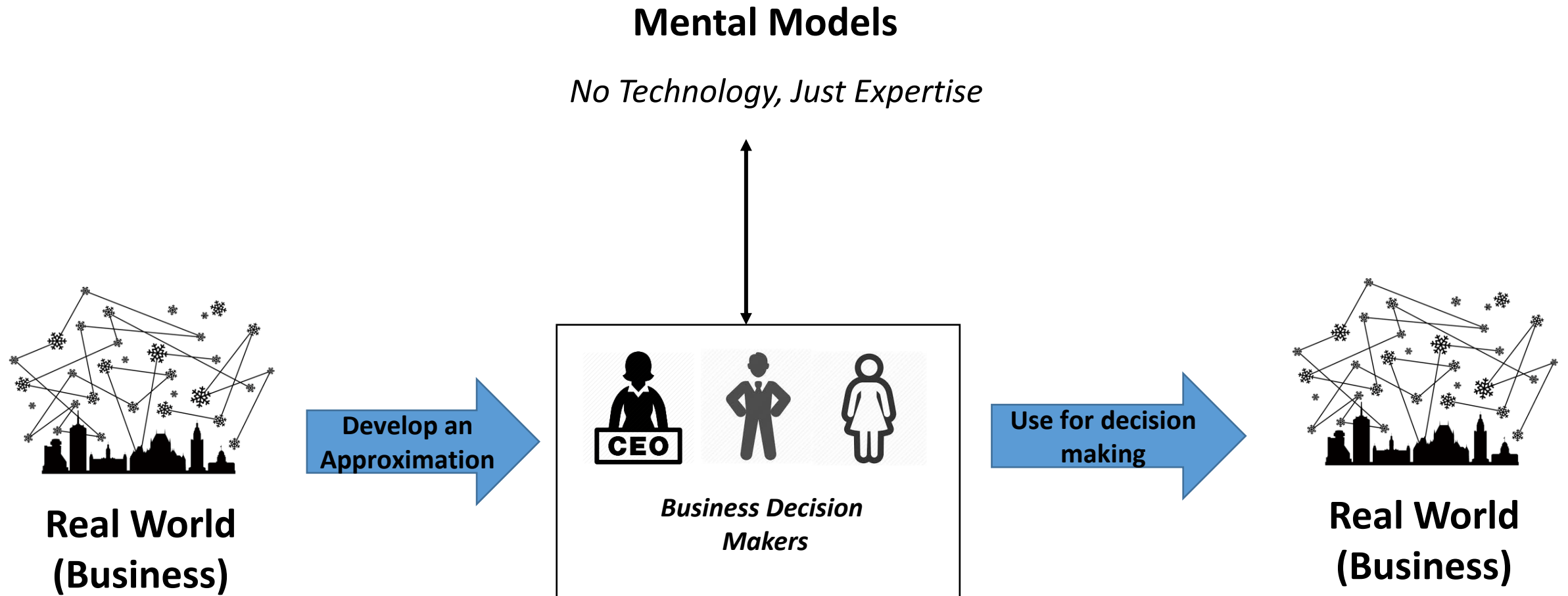
# Business Decisions – Type 3 - Operational



# So how are business decisions made?



# Mental Models backed by Human Expertise (Gut Feel)

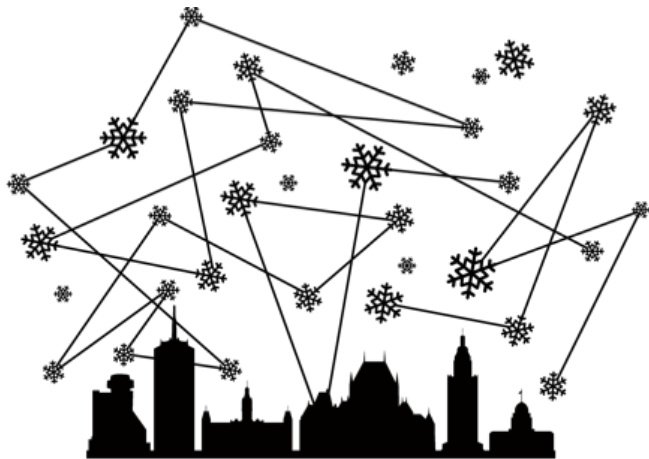


Q5: Decision making in a business context has been done for decades. Where does Analytics (AI/ML) play a role and why is there such a big interest now?



# Business Decision Making is becoming increasingly 'Complex'

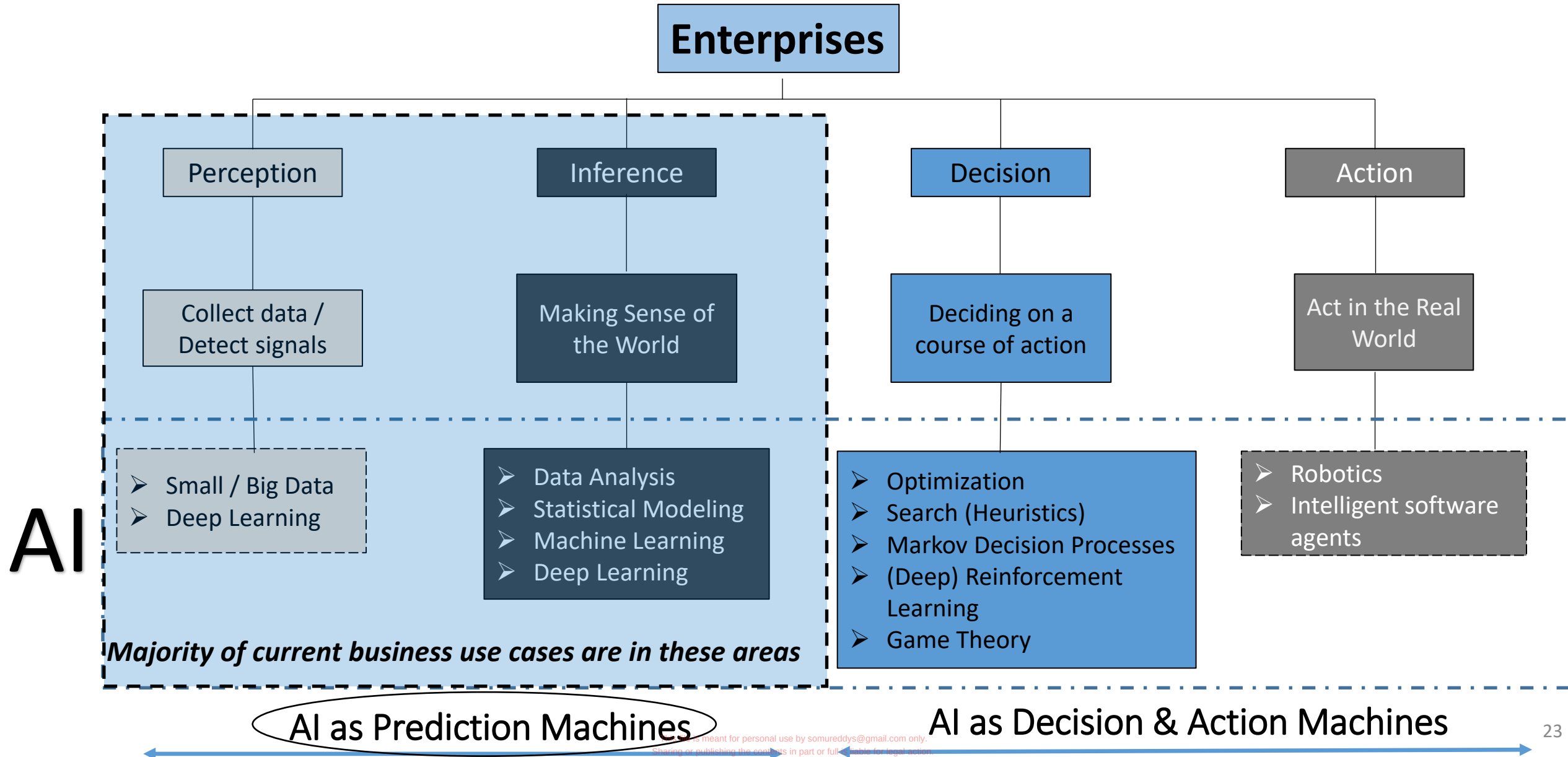
## Real World (Business)



- Multiple Entities with their own goals
- Complex Non-Linear Relationships
- Delayed Feedback
- 'Black Swan' Events (Unpredictable events)
- Uncertainty & Long Term Consequences

***Business decision makers are expected to take right decisions in this complex world***

# AI as Prediction Machines



# AI as Prediction Machines - Examples

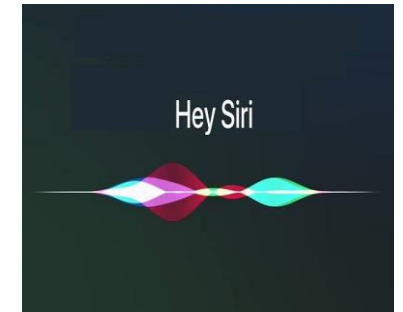
## *Prediction Machines*



Customer Attributes  
Product Details  
Transaction History

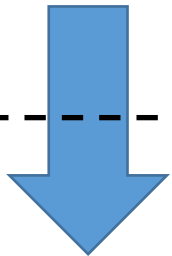


Product Sales  
across time  
Location of Stores



Utterance by a  
customer on his /  
her mobile phone

Take Information  
that you have  
(Data)



Create Information  
that you don't have  
(Predictions)

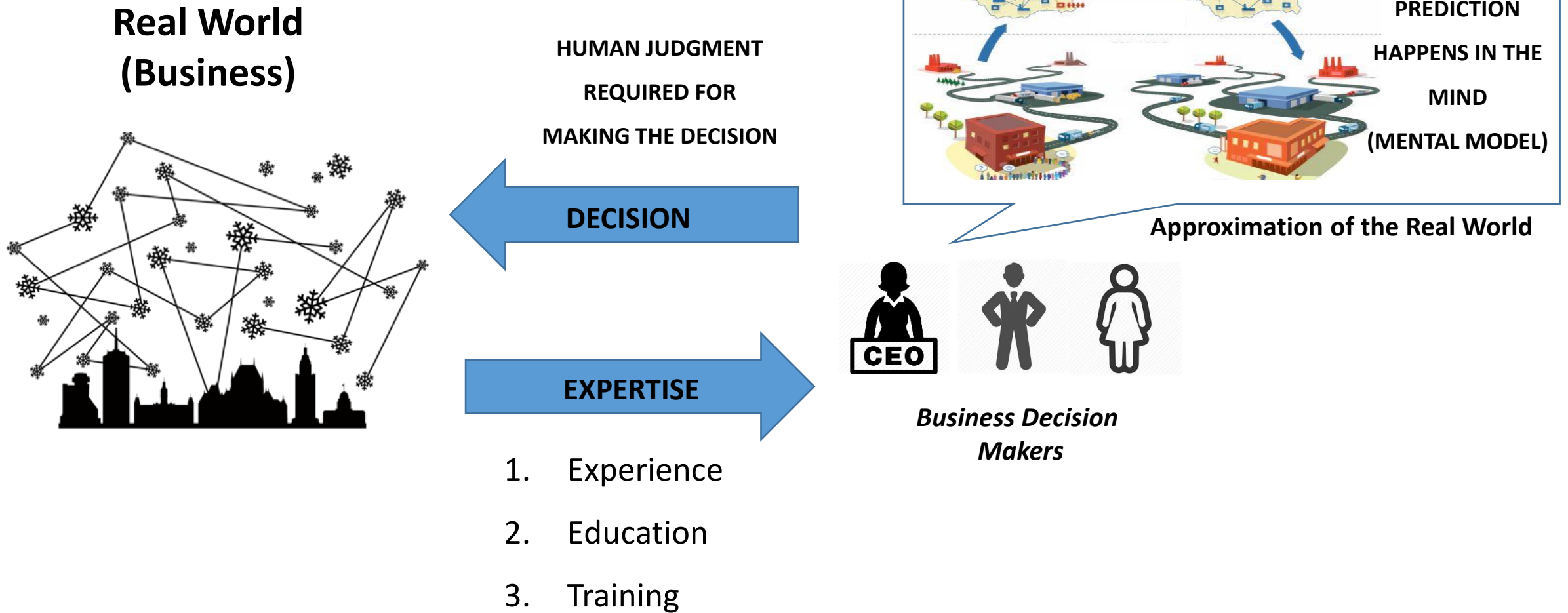
Whether a particular  
customer will churn  
(leave the bank) or  
not?

Forecast demand for  
products in the next 3  
months in each store

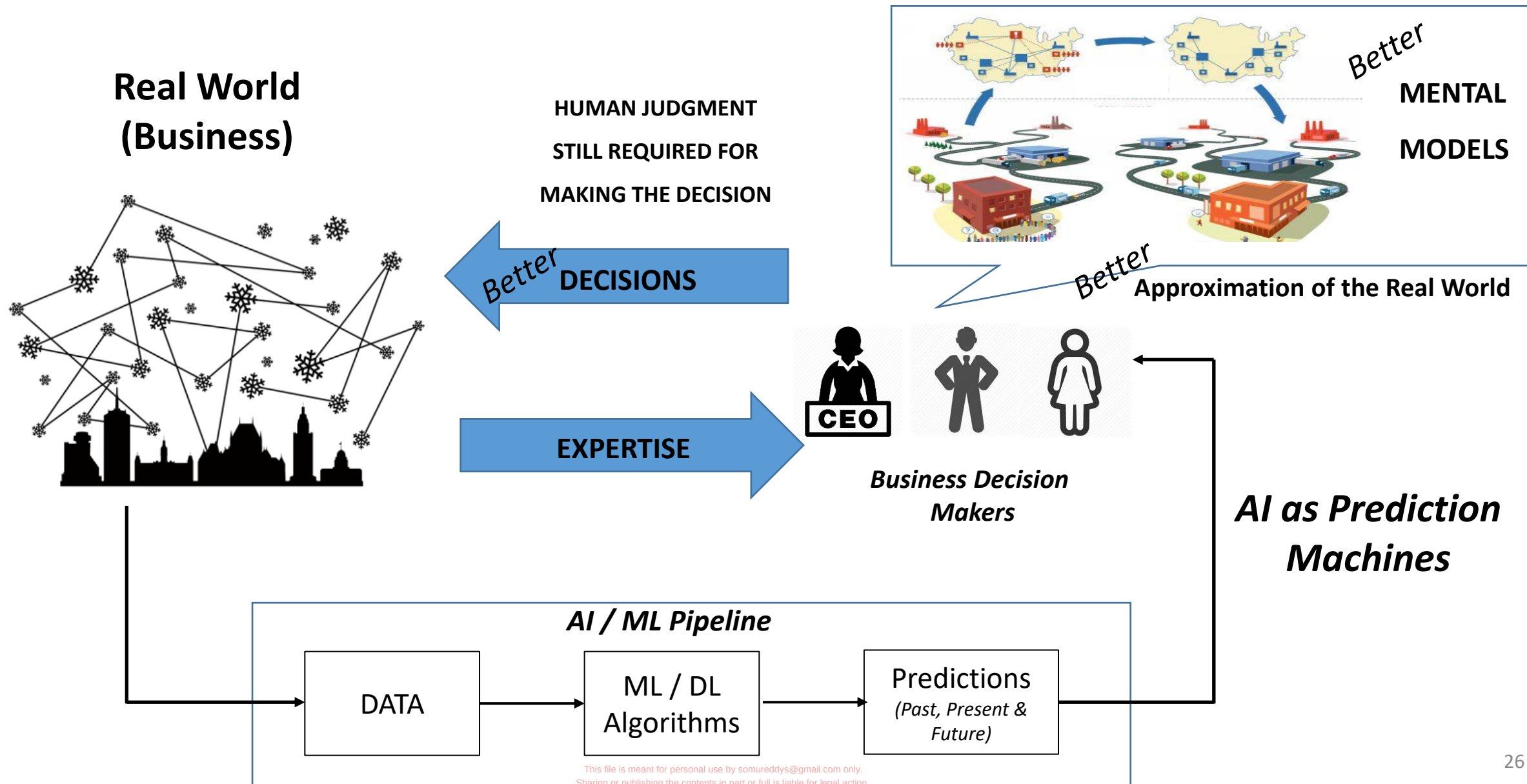
Precisely identify  
what the customer  
said so that action  
can be performed



# We saw earlier how business decisions are made?

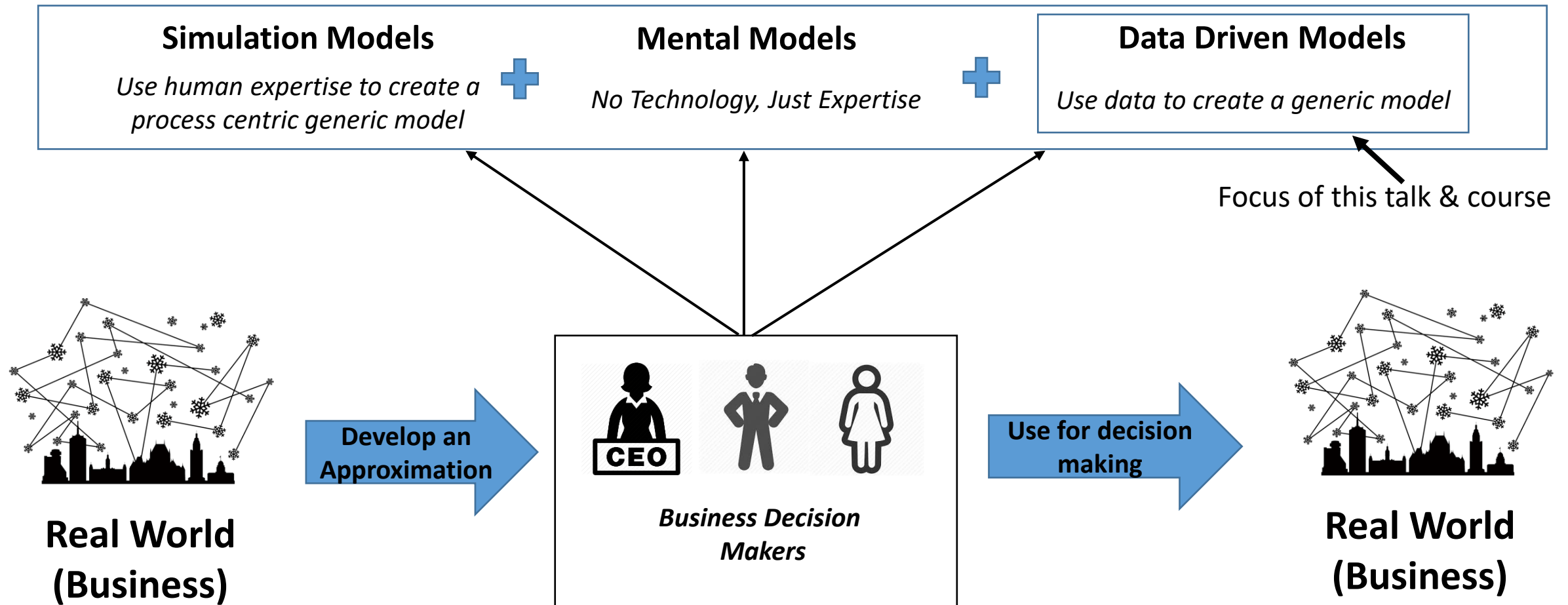


# Goal & Purpose of Analytics – (with AI as Prediction Machines)

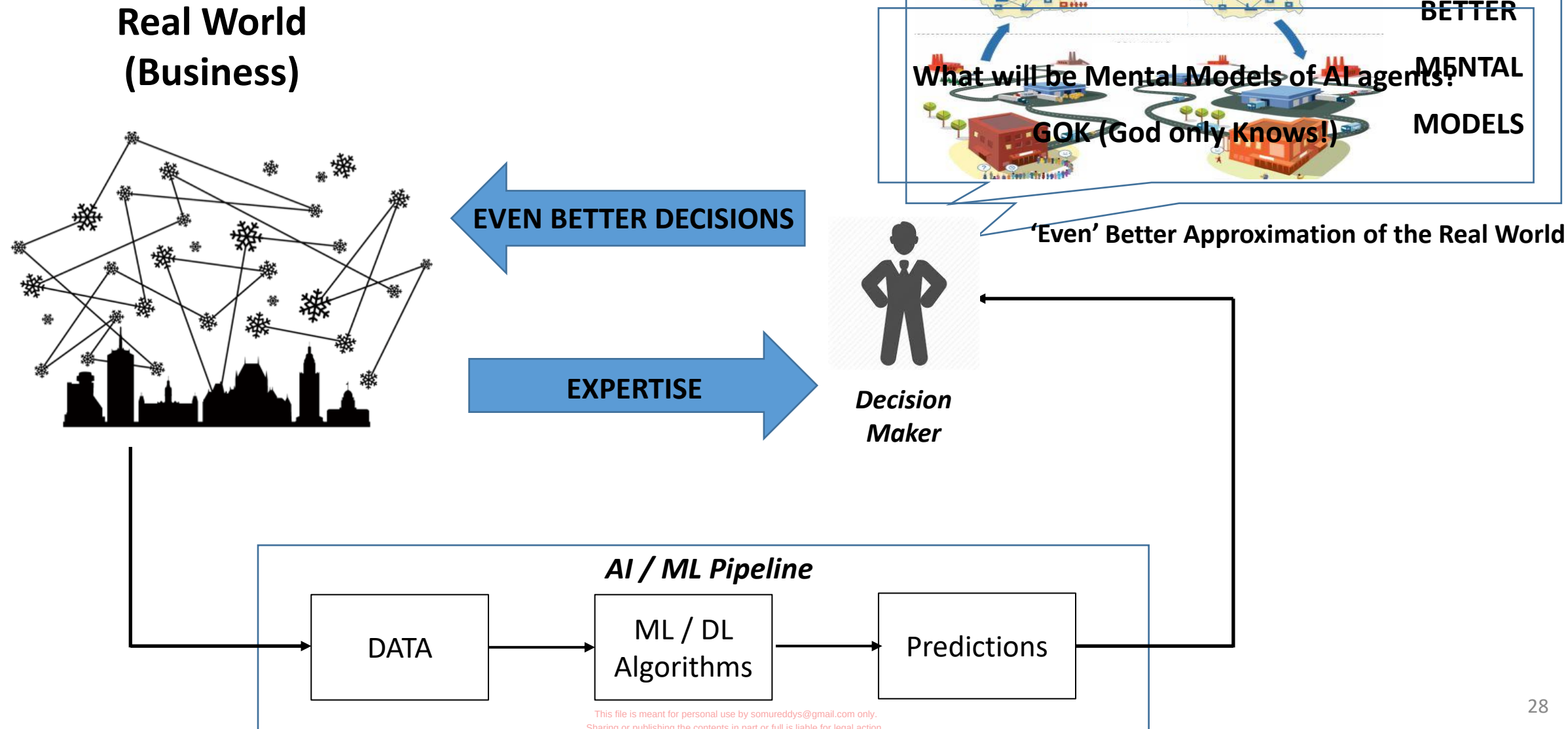


# 'Models of the Real World' is at the centre of business decisions

## Models of the Real World



# Artificial Intelligence (End to End / Autonomous)



Q6: Can you bring these concepts to life with some examples from your personal experience?



# Example 1: Automobiles – Utilizing Sensor data to predict defects



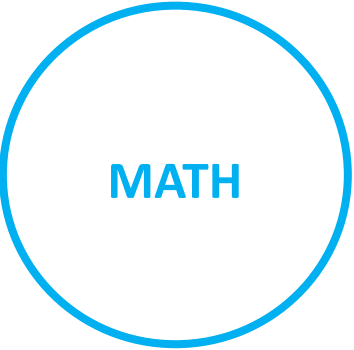
## BUSINESS

For this automobile company, the warranty costs were rising year on year at an alarming rate



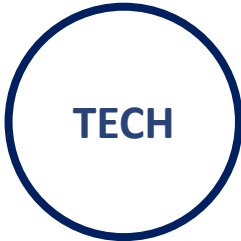
## DATA

Semi-structured data from sensors from cars across 100+ countries  
(Ex: Pedal position, Oil temperature, Engine temperature...60+ parameters)



## MATH

Clustering done on data to identify driving styles which is then correlated with warranty claims to predict defect probability



## TECH

- Spark on the Cloud platform called Databricks for Machine Learning
- User Interface using React for self-service



# Example 1: Utilizing sensor data to predict defects

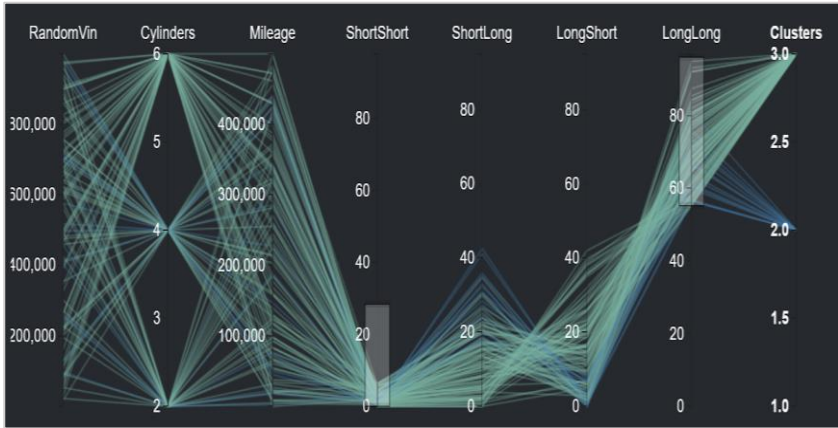
## AI as Prediction Machine

### Business Problem

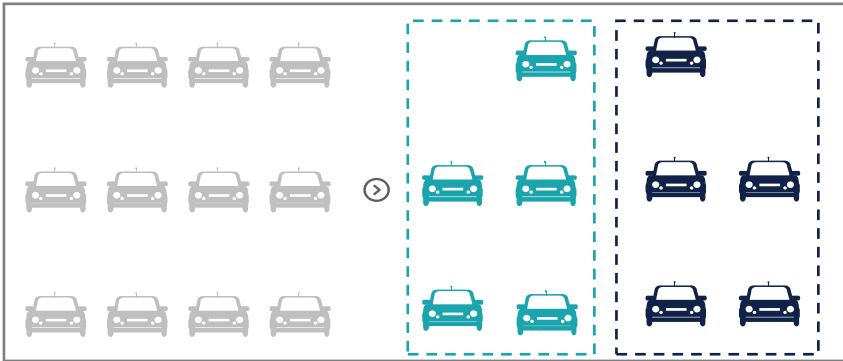
For this automobile company, the warranty costs were rising year on year at an alarming rate



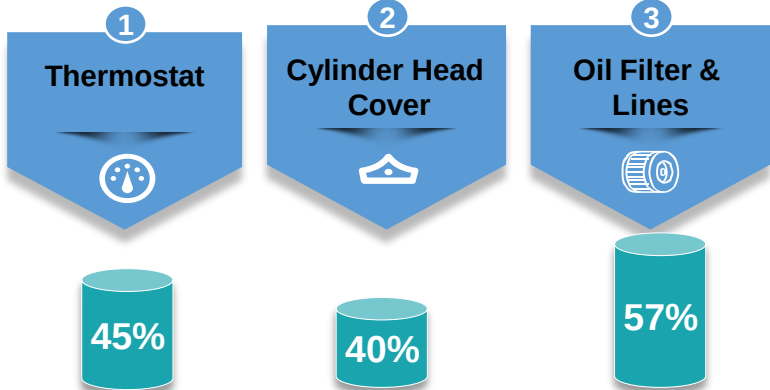
Sensor data from cars in 100+ different countries



Visualization to understand the data

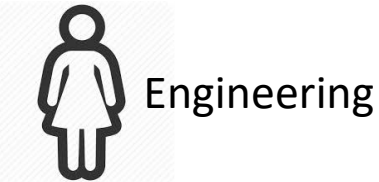


Clustering to identify driving styles



Defects were correlated with driving styles

### Decisions

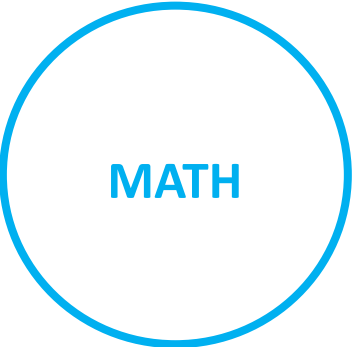


## Example 2: Leveraging external data to drive innovation



### BUSINESS

Consumer durables company wants to obtain product feedback as soon as its products are released in the market and not wait for 6-8 months which was the current state scenario?



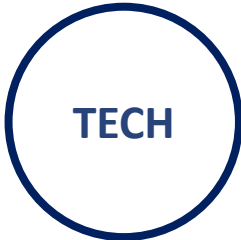
### MATH

Natural Language Processing  
Techniques to detect spam, emotion, entities, sentiments, contextual meaning etc.



### DATA

Unstructured data from reviews in marketplaces, brand websites, Industry forums, blogs



### TECH

Self-service visualization built using Tableau that provides the summary view and different levels of drill-down into specific consumer characteristics



# Example 2: Using External Data to Drive Innovation

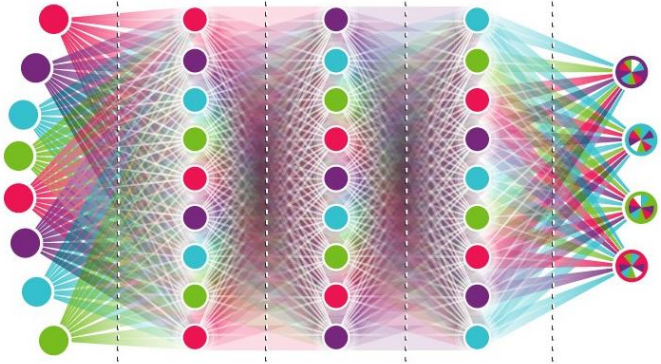
## AI as Prediction Machine

### Business Problem

Consumer durables company gets product feedback 6-8 months after the product is launched which is not useful



External data from ecommerce sites, brand websites etc.



Deep Learning for Text Understanding

### Decisions



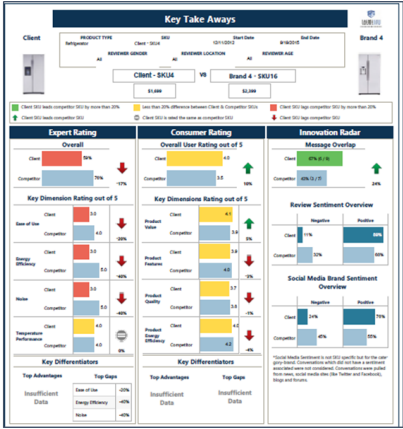
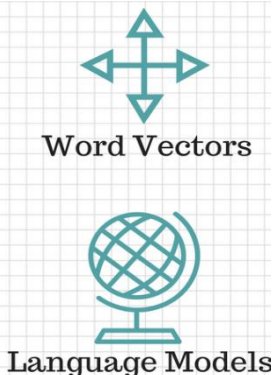
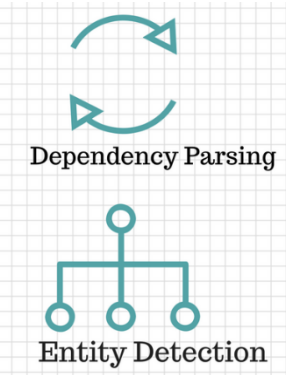
Innovation



Marketing



Distribution



Understand key factors that influence customer perception & behavior

Q7. What competencies are required to create data driven models?



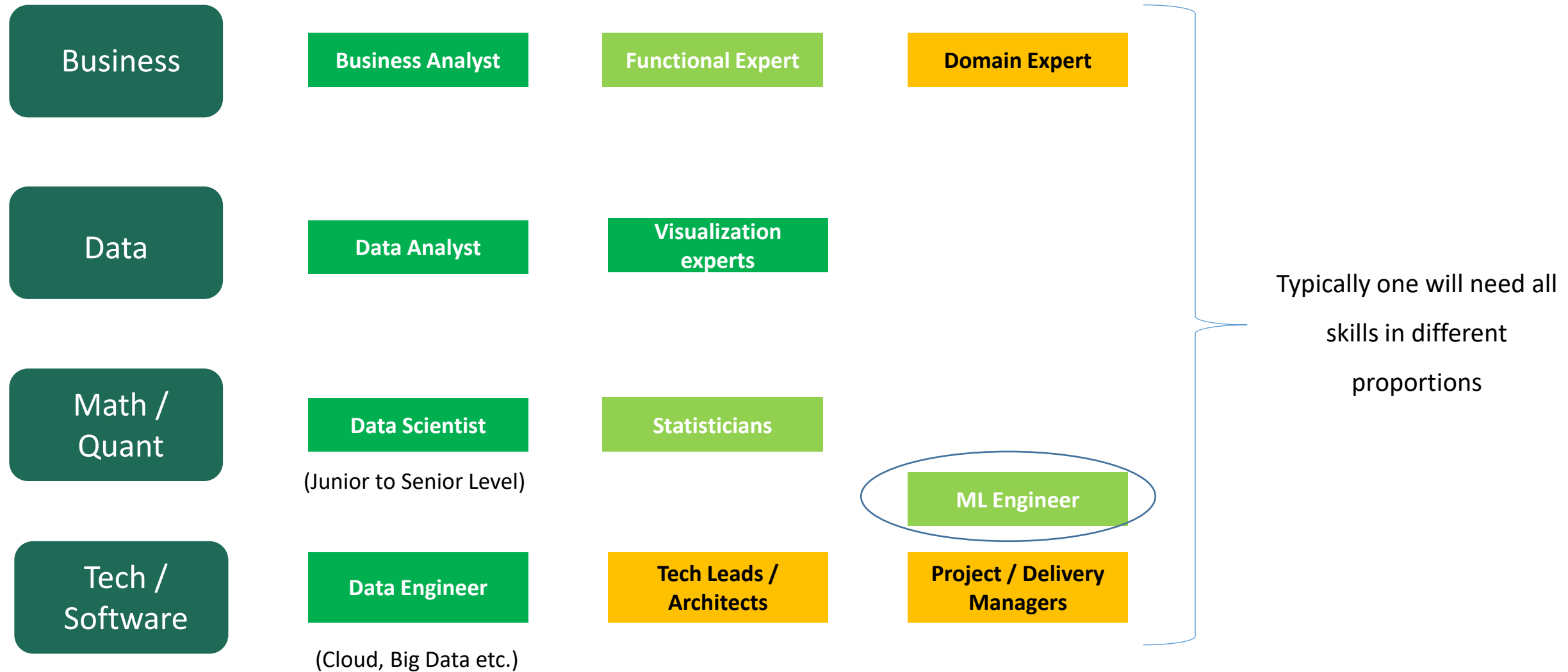
# Competencies to create Data Driven Models

|                 |                                     |                                        |                                    |
|-----------------|-------------------------------------|----------------------------------------|------------------------------------|
| Business        | Use Case Formulation                | Interpret Analytics Output             | Domain Expertise                   |
| Data            | Data Types – (Un / Semi) Structured | Data Visualization & Story Telling     | Signals from data (subtract noise) |
| Math / Quant    | Select the right techniques         | Appropriate coding language / platform | Evaluating the output of algos     |
| Tech / Software | Data Engineering / Model Deployment | Front-end Applications                 | Software Engineering / SDLC        |

Q8: What are the typical roles in the analytics space and what specific skills are required to break into it?



# Typical Roles in Analytics / Data Science



# Entry Points for different experience levels

| Fresher / Junior Developer  | Lead / Architect                            | Program Manager / Business Head |
|-----------------------------|---------------------------------------------|---------------------------------|
| Business Analyst            | Functional / Domain Lead                    | Functional / Domain Expert      |
| Data Analyst                | Senior Data Analyst                         | Mid-Level Data Scientist        |
| Big Data / Cloud programmer | Mid-Level Data Scientist                    | Analytics Delivery Manager      |
|                             | Data Engineer / Big Data / Cloud Specialist | Consultant                      |

# Skills required for an Analytics Professional

- 1. Business Orientation** – Basic understanding of business concepts
- 2. Structured Thinking** – Ability to break down a problem into parts
- 3. Analytical Techniques** – Articulate as to how data & analytics can help

# 1. Business Orientation - Basic understanding of business concepts

**Value Creation** – Reason for existence of a business

**Business Model** – Core actors & activities that generate value

Raw material /  
Ideas



Making a Product /  
Offering a Service

Marketing & Selling  
a Product / Service



Customers

**Finance** – What numbers matter & why?

**Strategy** – To deal with competition

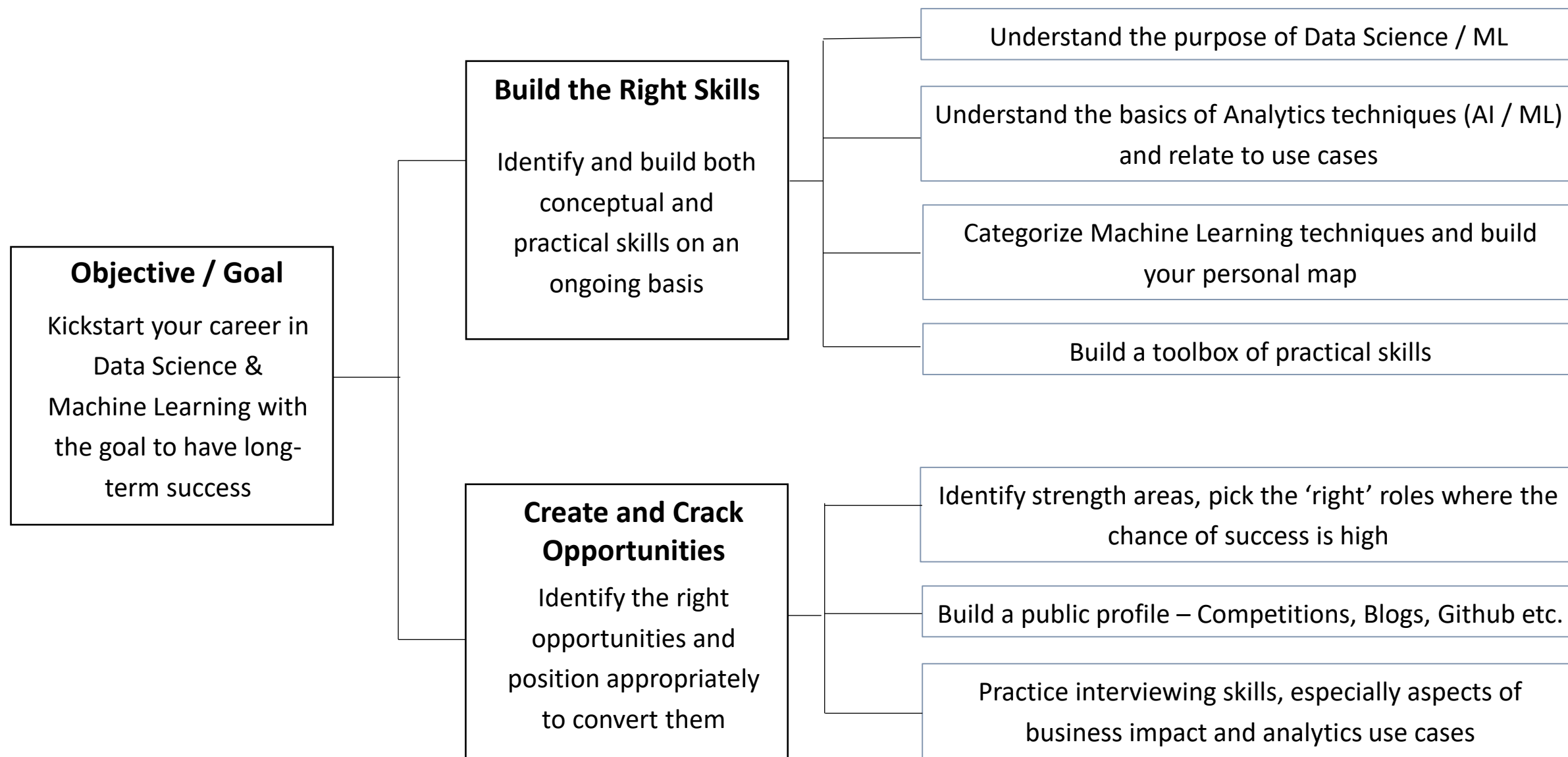
**R & D** – Innovation

**HR** – Managing People

Supporting  
Functions



## 2. Structured Thinking – Ability to break down a problem



# Structured Thinking - Many Resources are available...

## Frameworks

- Cynefin (by Dave Snowden)
- Issue Tree & Hypothesis Formulation
- MECE Principle
- Porter's Five Forces
- 3C Framework (Company, Customer & Competitor)
- 4P Framework (Product, Price, Place, Promotion)
- SWOT
- 2x2 Matrices
- BCG Matrix
- McKinsey 7S Framework

## Where Available

- Case Interview preparation websites like:
  - [www.craftingcases.com](http://www.craftingcases.com)
  - [www.preplounge.com](http://www.preplounge.com)
  - Victor Cheng's CaseInterview prep
- Consulting company websites  
(Ex: <http://caseinterviewprep.deloitte.com/>)
- YouTube (Case Interview Preparation videos)
- Forums like Quora

### 3. Analytics Techniques – Components of the toolbox

- Basic knowledge of Math concepts – Distributions, Probability, Matrices
  - Good knowledge of Statistical Techniques & Machine Learning algorithms
  - Atleast 1 programming language for ML – Python (Jupyter notebooks), R
  - Data Visualization skills – Python or Tools like Tableau, Qlikview
  - Atleast 1 GUI based ML platform – H2o, Azure ML, BigML
- 
- 1 Cloud based platform (Nice to have) – AWS, Databricks, Paperspace
  - Github
  - Kaggle (Competition & Kernels), AnalyticsVidhya
  - Database / SQL knowledge (preferable)

→ *Structured course will help you get this knowledge faster*

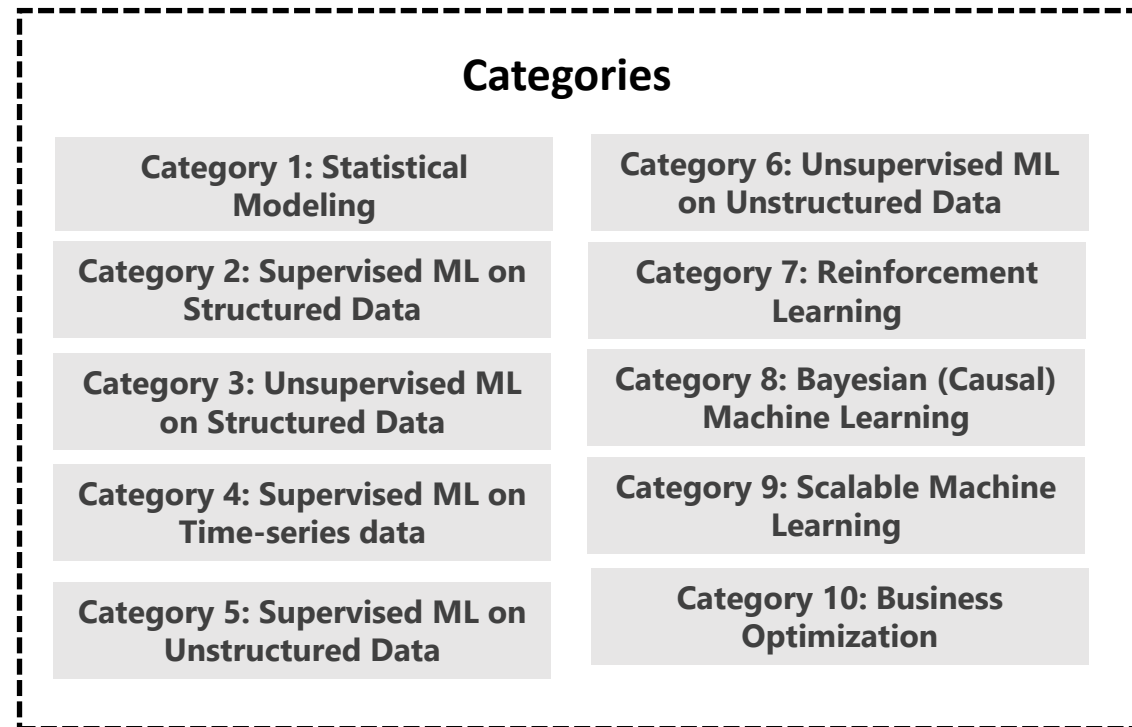
Q9: With so many algorithms & techniques available, how does one keep track and apply it?



# AI / ML Techniques – There are a lot of them!



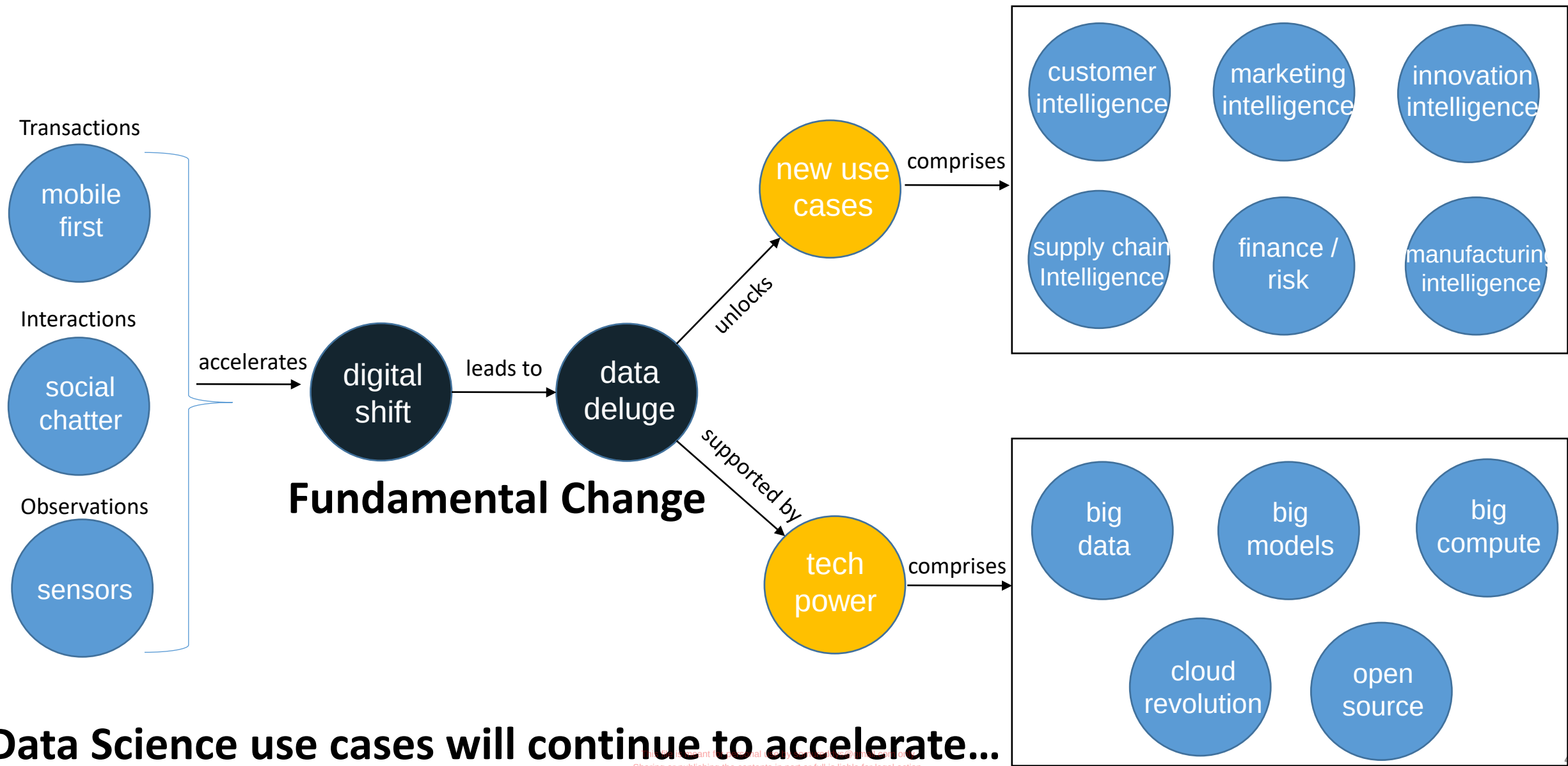
# Categorization of Machine Learning Topics – My personal map



Q10: How are you sure that this field (Analytics, AI & ML) has long-term growth prospects and is not just a short-term fad?



# Digital Shift is a Fundamental, Irreversible Change





# Thank You!



- Karthikeyan Sankaran, Director, LatentView Analytics
- LinkedIn – <http://in.linkedin.com/in/karthikeyansankaran>
- Mindmap – [bit.ly/31KArT8](http://bit.ly/31KArT8)